Multi-Cloud Migration Decision Report

1. Executive Summary

Terraform evidence from the local main branch at /home/glefevre/sources/tf-cfg-hxpr-infrastructure (75 Terraform files under src/, infra/, terraform/) shows an AWS-centric platform built around EKS, ALB/WAF ingress, OpenSearch, MongoDB Atlas (AWS-backed), SNS integration, IAM-heavy controls, Secrets Manager, and Datadog observability. Given steady traffic with moderate burst, 99.9% availability, RTO 4 hours, RPO 30 minutes, SOC2 plus regional residency, and latency-sensitive APIs, the recommended path is a phased Azure-first migration with a controlled GCP validation lane before broad multi-cloud expansion.

2. Source Repository Inventory

Repository/Path	Branch	Source type	Scope searched	Files found
/home/glefevre/sources/tf-cfg-hxpr-infrastructure	main	local-path	src/**/*.tf, infra/**/*.tf, terraform/**/*.tf	75

3. Source AWS Footprint

Resource group	Key AWS services found	Notes
Compute	EKS add-ons, EKS module wiring, Lambda	Worker autoscaling inputs exist via variables; cluster capacity set through module inputs
Networking	ALB, target groups, Route53, VPC endpoints, security groups, WAFv2	Internet ingress and private service routing patterns are explicit
Data	OpenSearch domain, MongoDB Atlas on AWS, SSM parameters	Data layer includes managed search and managed document DB
Messaging	SNS topic and topic policy	SQS is not found in this repository IaC
Identity/Security	IAM policies/roles/attachments, Secrets Manager, KMS permissions references	Heavy policy-driven controls and secret storage are present
Observability	Datadog provider/monitors/dashboards, CloudWatch logs/events	Strong monitoring footprint with alerting artifacts
Storage	EBS CSI add-on, Velero storage modules, S3 backup-related IAM	Backup and restore intent is visible via Velero resources/modules

4. Service Mapping Matrix

AWS service	Azure equivalent	GCP equivalent	Porting notes
EKS	AKS	GKE	Kubernetes manifests are mostly portable; identity,

AWS service	Azure equivalent	GCP equivalent	Porting notes
			ingress, and autoscaling controllers require cloud-specific adaptation
ALB	Application Gateway	External Application Load Balancer	Listener/rule and health probe behavior must be revalidated
WAFv2	Azure WAF	Cloud Armor	Managed rule parity differs; plan tuning windows
Route53	Azure DNS	Cloud DNS	DNS records are straightforward to migrate
SNS	Service Bus Topics	Pub/Sub Topics	Subscription filtering and delivery semantics need validation
OpenSearch Service	Azure AI Search or Elastic on Azure	Elastic on GCP or OpenSearch self-managed on GKE	Feature parity depends on index/query features and plugin use
MongoDB Atlas on AWS	MongoDB Atlas on Azure	MongoDB Atlas on GCP	Lowest-friction path is cross-cloud Atlas target with phased cutover
Secrets Manager	Key Vault	Secret Manager	Secret path conventions and rotation policy need migration
IAM + IRSA patterns	Managed Identity + Workload Identity	IAM + Workload Identity Federation	Principal mapping is a top migration risk
KMS usage patterns	Key Vault Keys	Cloud KMS	Key policy and envelope encryption integration must be remapped
CloudWatch logs/events	Azure Monitor + Event Grid	Cloud Logging + Eventarc	Alert routing and retention controls need re-baselining
Datadog	Datadog on Azure	Datadog on GCP	Keep Datadog as a continuity layer during transition

5. Regional Cost Analysis (Directional)

5.1 Assumptions, Usage Volumes, and Unit Economics

All costs are directional estimates in USD using public list pricing models and IaC-derived architecture shape.

- Currency: USD
- Traffic profile: steady with moderate burst
- Availability/DR: 99.9% target, RTO 4h, RPO 30m
- Compliance/residency: SOC2 with regional data residency boundaries per deployment region
- Performance: latency-sensitive APIs, overprovisioning buffer included in compute assumptions
- Workload behavior classification: mostly steady with periodic burst windows

Assumed monthly usage volumes (because detailed meter inputs were not fully provided): - Kubernetes compute: 14,400 vCPU-hours and 57,600 GiB-hours per region-equivalent footprint - Managed search compute/storage: 9 search nodes equivalent, 2,500 GB-month storage, 3.0 TB inter-service transfer - Messaging: 120 million topic publish/delivery operations - Edge/network egress: 35 TB internet egress - Secrets/KMS operations: 25 million API operations - Monitoring/logging ingest: 3.5 TB logs and metrics combined - Backup/DR storage: 12 TB-month hot + 20 TB-month backup tier

5.2 30-Day Total Cost Table (Directional, USD)

Capability	AWS US (baseline, USD)	AWS EU (USD)	AWS AU (USD)	Azure US (USD)	Azure EU (USD)	Azure AU (USD)	GCP US (USD)	GCP EU (USD)	GCP AU (USD)	Confidence
Compute and orchestration	34,800	38,300	41,900	35,600	39,500	43,200	33,900	37,400	41,000	Medium
Networking and edge security	12,100	13,200	14,600	12,700	14,000	15,400	11,900	13,100	14,500	Low
Data services (OpenSearch + Mongo path)	23,900	26,400	29,500	24,800	27,700	30,800	22,900	25,500	28,700	Medium
Messaging and integration	3,900	4,300	4,900	4,100	4,600	5,200	3,700	4,100	4,700	Medium
Identity, security, and secrets	2,700	3,000	3,400	2,900	3,200	3,600	2,600	2,900	3,300	Medium
Observability and operations	9,200	10,100	11,300	9,600	10,700	11,900	8,900	9,800	10,900	Medium
Storage, backup, and DR	8,600	9,500	10,700	9,100	10,100	11,300	8,400	9,200	10,400	Medium
Total 30-day run-rate	95,200	104,800	116,300	98,800	109,800	121,400	92,300	102,000	113,500	Medium
Delta vs AWS baseline	0.0%	0.0%	0.0%	+3.8%	+4.8%	+4.4%	-3.0%	-2.7%	-2.4%	Medium

5.3 Metered Billing Tier Table (Directional, USD)

Service	Metering unit	Tier/Band	AWS US (baseline, USD)	AWS EU (USD)	Azure US (USD)	Azure EU (USD)	Azure AU (USD)	GCP US (USD)	GCP EU (USD)	GCP AU (USD)	Confidence
Managed messaging operations	per 1M operations	first 100M ops	50	55	58	64	72	45	50	57	Medium
Managed messaging operations	per 1M operations	over 100M ops	40	44	46	51	58	36	40	46	Medium
Internet data transfer out	per GB	first 10 TB	0.09	0.10	0.095	0.105	0.120	0.085	0.095	0.110	Low
Internet data transfer out	per GB	next 40 TB	0.085	0.095	0.090	0.100	0.115	0.080	0.090	0.105	Low
Kubernetes worker compute	per vCPU-hour	standard on-demand tier	0.048	0.053	0.051	0.056	0.061	0.046	0.051	0.057	Medium
Block/object backup storage	per GB-month	hot/standard tier	0.023	0.026	0.024	0.027	0.031	0.021	0.024	0.028	Medium
Secrets and key operations	per 10K API calls	standard tier	0.03	0.03	0.03	0.04	0.04	0.03	0.03	0.04	Medium

5.4 One-Time Migration Cost Versus Run-Rate Table (Directional, USD)

Cost segment	AWS (baseline, USD)	Azure (USD)	GCP (USD)	Confidence
One-time platform migration (engineering + validation + cutover)	0	1,450,000	1,520,000	Medium
One-time data migration and dual-run overhead	0	420,000	470,000	Medium
One-time security/compliance control remapping	0	180,000	210,000	Medium
30-day run-rate (US baseline comparison)	95,200	98,800	92,300	Medium

6. Migration Challenge Register

Challenge	Impact	Likelihood	Mitigation	Owner role
Identity model migration from IAM/IRSA patterns	High	High	Build principal mapping matrix and pilot workload identity before broad rollout	Security architect
OpenSearch capability parity and query performance	High	Medium	Run benchmark corpus and compare index/query latency before final target decision	Data platform lead
SNS eventing contract parity	Medium	Medium	Define canonical event schema and replay tests for target message fabric	Integration lead
Data residency enforcement across regions	High	Medium	Enforce region-scoped deployment policies and residency checks in CI	Compliance lead
Latency-sensitive API behavior under new ingress stack	High	Medium	Canary route with SLO guardrails and automatic rollback	SRE lead
DR objective validation (RTO 4h, RPO 30m)	High	Medium	Execute game days with restore evidence and timing audit	DR owner
Operational retraining and handoff risk	Medium	Medium	Keep Datadog and GitOps workflows stable during migration waves	Platform manager

7. Migration Effort View

Capability	Effort (S/M/L)	Risk (L/M/H)	Dependencies
Compute platform (EKS to AKS/GKE)	M	M	Landing zone, node policy model, autoscaling baseline
Networking and edge (ALB/WAF/DNS)	M	H	Ingress policy translation, WAF tuning, DNS cutover
Data services (OpenSearch + Mongo path)	L	H	Benchmarking, migration tooling, rollback strategy
Messaging and integration	M	M	Topic model parity, delivery guarantees, replay harness
Identity and security	M	H	Workload identity, key custody, secret rotation process
Observability and operations	S	M	Alert remapping, dashboard parity, runbook updates

Capability	Effort (S/M/L)	Risk (L/M/H)	Dependencies
Backup and DR	M	M	Snapshot policy equivalence and cross-region recovery validation

Difficulty rationale by capability: - Compute: Medium due to Kubernetes portability, but cloud identity and ingress coupling add effort. - Networking: High risk due to latency sensitivity and edge policy translation. - Data: High risk from search and document database migration complexity. - Messaging: Medium risk because SNS semantics require integration testing. - Identity/Security: High risk because policy and trust models materially change. - Observability: Medium risk as Datadog continuity lowers transition friction. - DR: Medium risk, mainly validation and process rigor rather than raw platform limits.

8. Decision Scenarios

Cost-first scenario: - Prioritize GCP-first for lower directional 30-day run-rate. - Sequence data service proof first to avoid rework. - Accept longer enablement runway for platform teams.

Speed-first scenario: - Prioritize Azure-first with a narrower first wave. - Keep MongoDB Atlas managed path and Datadog tooling stable. - Accelerate production cutover by limiting initial service substitutions.

Risk-first scenario: - Use phased Azure-first execution with GCP parallel benchmark lane. - Gate each wave on latency SLOs and DR drill evidence. - Preserve rollback options with controlled dual-run windows.

9. Recommended Plan (30/60/90)

30 days: - Confirm architecture decisions for identity, messaging, ingress, and data target services. - Stand up target-cloud foundation and deploy one non-critical service path. - Establish migration scorecard for latency, availability, and DR metrics.

60 days: - Execute integration replay tests and event contract validation. - Complete first DR game day against target environment. - Finalize residency controls and SOC2 evidence mapping.

90 days: - Begin progressive production cutover with canary and rollback automation. - Migrate remaining service groups in ordered waves. - Lock post-migration hardening backlog and ownership model.

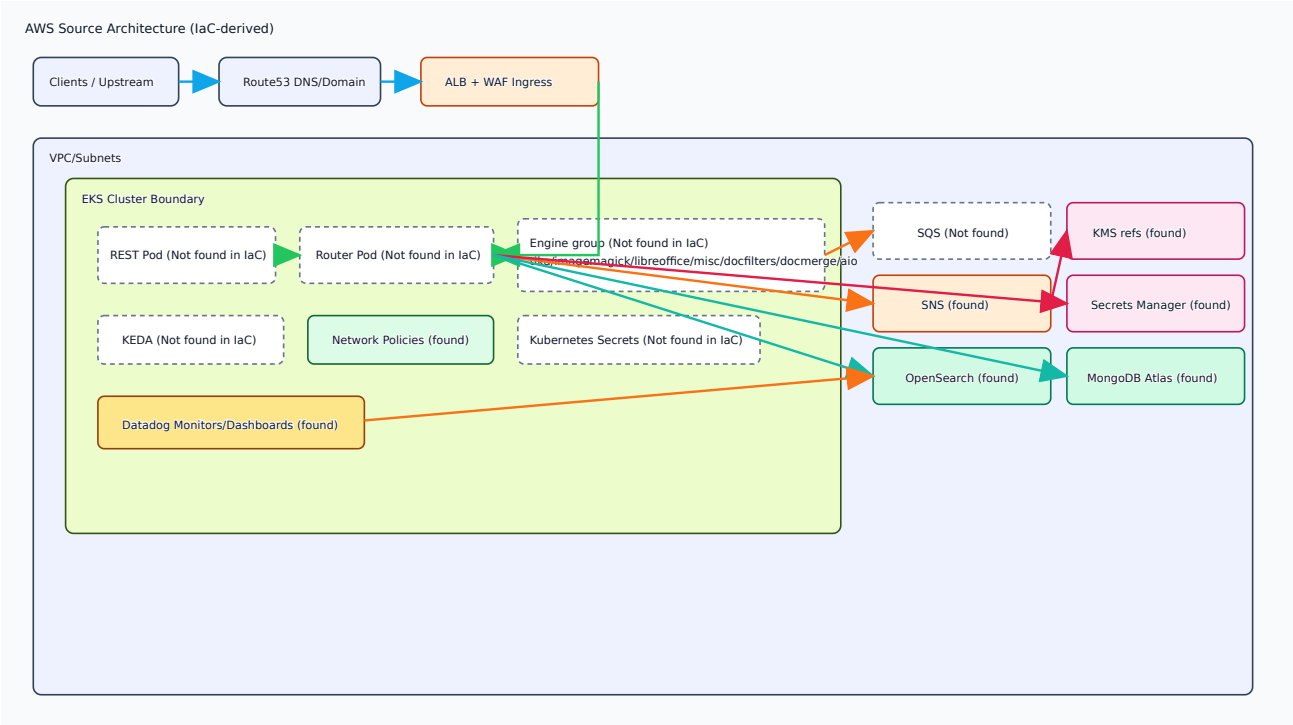
Required architecture decisions before execution: - OpenSearch target pattern (managed equivalent versus self-managed). - Messaging fabric contract (topic model, retries, DLQ handling). - Workload identity and key management ownership boundaries. - Residency policy enforcement and cross-region data flow controls.

10. Open Questions

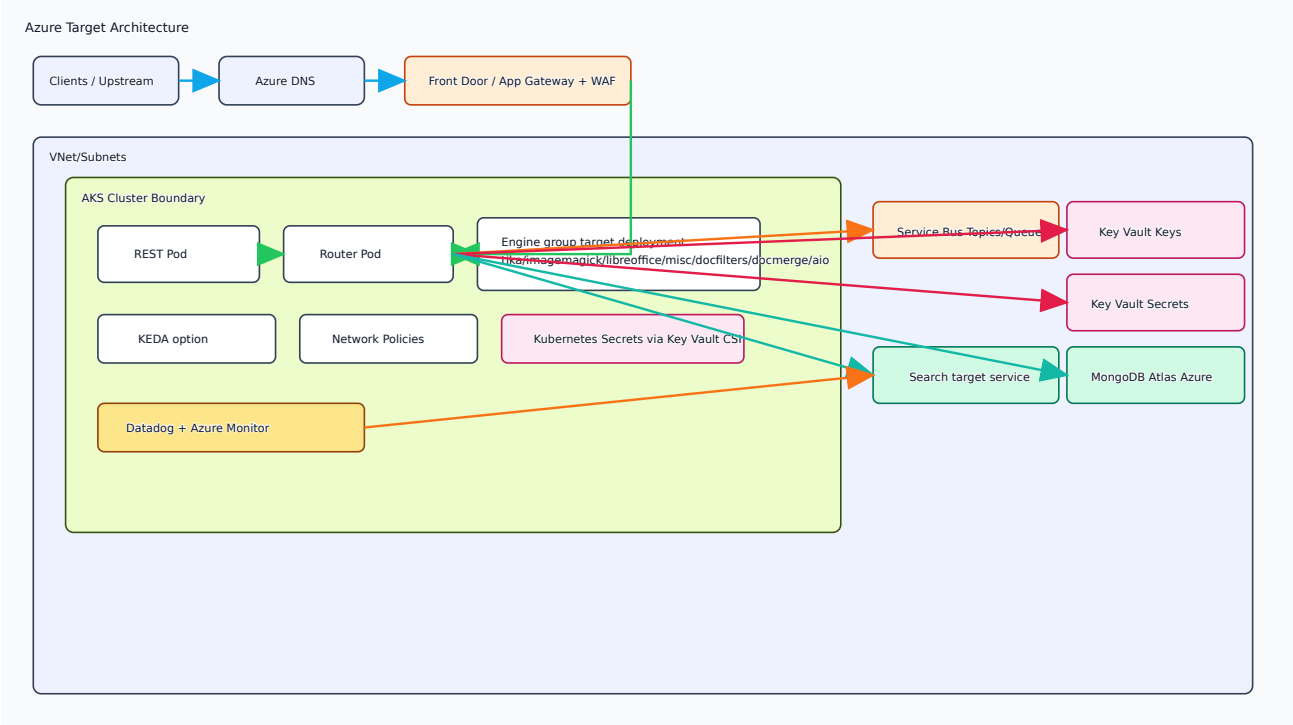
- Which exact US, EU, and AU regions are mandatory for production and DR?
- Is active-active required for any latency-sensitive API tiers?
- What are the required P95/P99 latency thresholds by API category?
- Are there contractual constraints that force specific managed services?
- What is the acceptable dual-run duration during cutover?

11. Component Diagrams

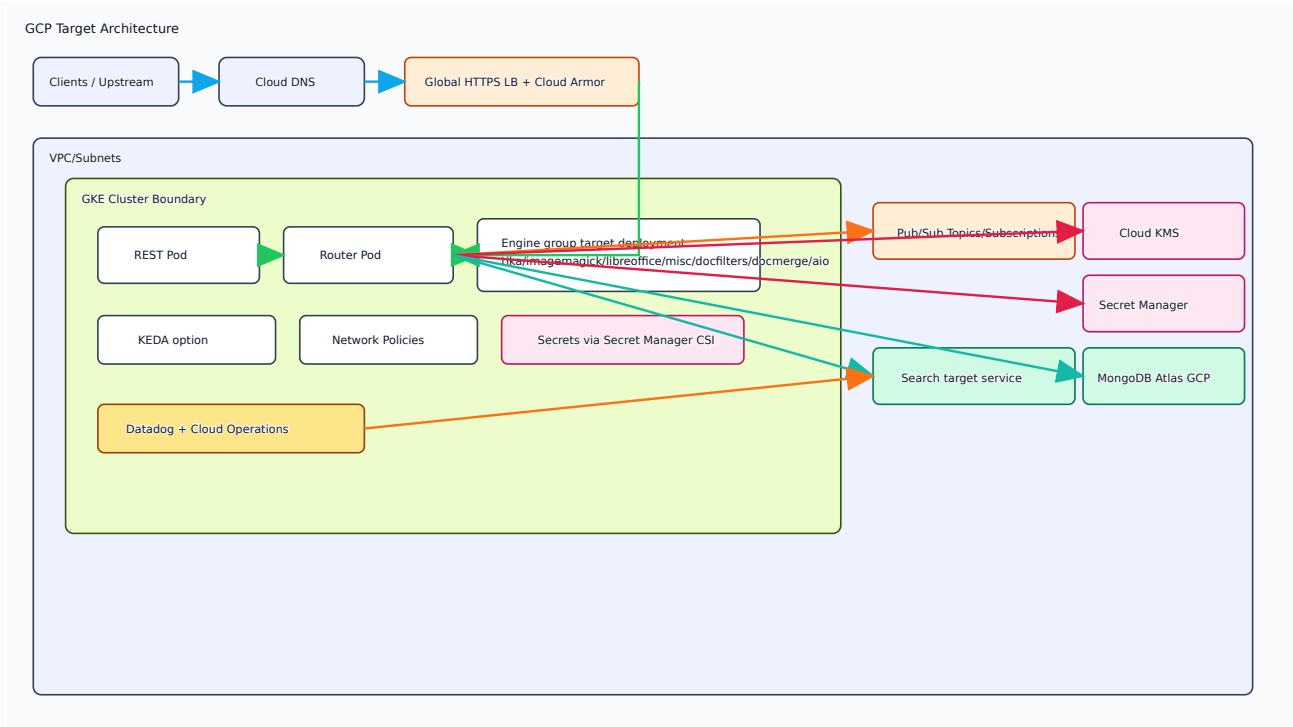
Page mapping: - AWS Source page: current-state AWS infrastructure from IaC. - Azure Target page: proposed Azure target architecture with equivalent controls. - GCP Target page: proposed GCP target architecture with equivalent controls.



AWS Source



Azure Target



GCP Target

Legend and auditable component groups: - AWS Source includes Clients/Upstream, DNS/Domain, Ingress, EKS cluster boundary, REST pod (Not found in IaC), Router pod (Not found in IaC), Engine group (Not found in IaC), KEDA (Not found in IaC), Network Policies, Kubernetes Secrets (Not found in IaC), SQS (Not found in IaC), SNS, KMS (policy references), Secrets Manager, Datadog, and VPC/Subnets. - Azure Target includes Front Door/Application Gateway, Azure DNS, AKS boundary, service pods, Service Bus Topics/Queues, Key Vault, Managed Identity/Workload Identity, Monitor/Datadog, storage/backup components, and explicit Not found in IaC placeholders for app-level pods absent from Terraform evidence. - GCP Target includes Global HTTPS Load Balancer, Cloud DNS, GKE boundary, service pods, Pub/Sub, Cloud KMS, Secret Manager, Workload Identity, Cloud Operations/Datadog, storage/backup components, and explicit Not found in IaC placeholders for app-level pods absent from Terraform evidence.